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A Reflective Journey: Navigating My Cumulative Experience at Iowa State University

Overall, Iowa State has been an instrumental stage of life for me to learn how to devise a plan, fail fast during problem-solving, and adapt to challenges quickly. My biggest lessons in engineering and life, in general, have been learned from the toughness in my academics and the soft skills I've had to develop to earn an engineering degree. In engineering, failing and trying again is not considered harmful but a necessary evil to learn harsh lessons that strengthen knowledge. Without this rigor and challenge, Iowa State provides in the engineering program, it would be a cakewalk that does not reflect the challenge of real-world problems engineers have to solve daily to improve people's lives. While I was frustrated when adversity was extremely high, I am grateful for these opportunities to observe what I can learn under pressure and become a wiser engineer, leader, and person.

Additionally, many classes discussed the societal and ethical implications of design choices. Almost every class touched on economical tradeoffs, such as performance versus energy cost, efficiency, accuracy of results, and material or other overheads. These concepts and options are essential to the engineering field, as business drives engineering solutions, and the economics of decisions drive how businesses interact with their engineering teams in requirements gathering, collaboration, and deliverables. In classes like CPR E 288 and CPR E 381, I learned the complex technical nature of how technologies like embedded systems and computer processors are designed, the design choices engineers make to balance efficiency and performance with cost and got to collaborate with lab partners to apply design concepts to a physical deliverable. I thoroughly enjoyed the application aspect of my engineering experience, as problem-solving relies on both theory and the application of theory in real-world solutions. I value experiences of designing, building, and testing working solutions to problems, which is the main reason I chose to pursue engineering in the first place. And in C E 388, we took a holistic approach to

ethics with how engineering solutions may work in one environment or culture and not in others. These classes, among others, gave me examples of how to approach problems by first considering who and what may be impacted by solutions and what tradeoffs exist between solutions, making me feel confident I can choose the best solution when the time comes.

What I found to be most successful in completing class projects and problem-solving tasks was taking inspiration and understanding from various resources, whether the provided lecture notes and textbook, external publications, blog posts, and forums, or even my classmates taking the class with me. Iowa State's library is one of the university's most underrated resources, with an astronomical collection of knowledge and information and access to all kinds of online resources that would otherwise require payment. There were many times I wanted to reference an additional text or use a scholarly source for a paper that the library gave me access to. Additionally, the ability to have like-minded students with me in my classes made for great experiences of collaboration and exchange of ideas, when permitted, of course, to come up with a definitive solution. It makes no sense to do challenging classes and labs alone. Engineers in the field share ideas and work together to solve some of the world's most complex problems, so why should we practice trying to solve class problems on our own for four years? I also developed great friendships with the students I was able to collaborate with and even found opportunities to work with them on new projects later on, having seen the skillsets they developed for themselves and how they could benefit a broader team.

Thanks to Iowa State's incredible career fair offerings, I had many opportunities to practice interacting and interviewing with recruiters. To gain a well-paying job after graduation, internship experience in industry as a student is vital. The career fairs and career preparation resources Iowa State provides were excellent for developing resumes and cover letters and practicing for interviews, both technical and non-technical. I had staff on campus to help me practice selling my skills and character to employers beforehand, so I felt prepared and empowered for the interaction. I gratefully received plenty of internship offers and accepted positions for two summers. I learned invaluable skills in applying my

degree knowledge to real-world business situations and how businesses, big and small, work with their customers, internal team members, and industries.

During one year, I participated in the College of Engineering's Make to Innovate program, allowing me to work on complex student-led aerospace projects. With one of the most challenging projects I had ever worked on, I collaborated with various student engineers from different departments and solved a history-making ISU project: we launched the university's first student satellite into space! The combination of intense technical challenges and understanding a multi-year and pages-long problem provided the incredible joy of a working system deployed successfully in the real world–a true capstone! A primary interest of mine I also got to pursue during my time, thanks to Iowa State's Pappajohn Center for Entrepreneurship, was starting my own software engineering business. I discovered a beautiful combination between solving an engineering challenge and using soft business skills to grow a company by thoroughly investigating a customer problem and leading a team to develop a profitable solution. And as another massive opportunity for me, I took in my internship experience and lessons and applied them to my own business. I had a medium to use my technical skills further while developing new soft skills like public speaking, business pitching, and leadership, all while positively impacting my community with our solution.

Finally, one of my favorite experiences that offered a surprising amount of learning opportunities was in the athletic bands. The marching band was an incredibly unique way to support the university I attend and expand my leadership skills by serving as a Drumline captain. In music, I got to use a different part of my brain to still collaborate with outstanding students and perform vibrant shows. I further had a way to apply leadership skills in my community by setting an example for fellow band members, handling logistics, and providing group support when necessary, skills that will carry into my next career stage.

With changing and growing industry conditions, they constantly demand engineers to learn new technologies to solve different problems. In my field, the solutions my industry creates evolve rapidly, sometimes faster than Moore's Law predicts. With the growth of topics like Artificial Intelligence and Big Data, new tools arrive every quarter, and old ones fade away in favor of more efficient ones. As an

engineer, my technical skills cannot stay static and instead must adjust for what jobs require of hires. For internships, I've had to learn languages like C#, tools like Jira, and strategies like Agile were not necessarily covered in my classes then. In internship development programs at companies, they preach to students that technical skills fade after 5 years in favor of new ones.

But what Iowa State has taught me in class is not one particular tool or technical skill, but how to apply a generic problem-solving approach to challenges I might face in industry. For hardware design, description languages like Verilog and VHDL might only be used by a few companies, but the skills I learned using those tools are just as transferable to those hardware design fields. The more tools and languages I know, the easier it is to understand the next one after being exposed to various syntaxes, behaviors, architectures, and design patterns. Iowa State does not offer "How to Program in X language" but rather broad concepts like computer organization, embedded systems, operating systems, etc., that can be transferred to any tools or demands the industry might have. This diversity in curriculum and problem-solving education is a leg-up I have for my entire career, and I am grateful for this approach.

Looking back at my experience, with my perspective on the other side of my four years, I would tell my freshman self to do it differently (but not by much). I would hit the ground sprinting in each class, immediately observing which parts of the material I struggle with and pouring extra time into answering my questions and uncertainties. This strategy would warm me to learn right at the beginning of the semester and expand my mind to other ideas and understandings far before I am expected to retain information for examination. Additionally, I would focus more on the learning experience, the journey I have to take through a course to learn the technical material, and the work ethic skills, as this is felt through most of the college experience. Commencement is only one day. The college learning experience is four years. While I think I understood this later in my experience, I would tell my freshman self to take it easy one day at a time and pause to reflect on how fantastic the learning process is, to find myself with new knowledge and skills I didn't have before. Regardless of what I wish I would have done differently, the skills, experiences, and knowledge I have at the end of my Iowa State journey are invaluable and will set me on a successful path I cannot wait to embark on.